

Quantifying measurement incompatibility of mutually unbiased bases

Supplementary Material

Sébastien Designolle,¹ Paul Skrzypczyk,² Florian Fröwis,¹ and Nicolas Brunner¹

¹*Département de Physique Appliquée, Université de Genève, 1211 Genève, Switzerland*

²*H.H. Wills Physics Laboratory, University of Bristol, Tyndall Avenue, Bristol, BS8 1TL, United Kingdom*

I. UPPER BOUND FOR ARBITRARY SETS OF POVMS

A. Dual problem

Here we derive the dual formulation of the SDP (4) of the main text in the general case of arbitrary POVMs. To that end, let us first write down the Lagrangian for this problem by introducing the Lagrange multipliers (dual variables) $\{Z_{\vec{j}}\}_{\vec{j}}$, $\{X_{a|x}\}_{a,x}$ and θ ,

$$\begin{aligned} \mathcal{L} = & \eta + \text{tr} \sum_{\vec{j}} Z_{\vec{j}} \mathcal{G}_{\vec{j}} + \theta(1 - \eta) \\ & - \text{tr} \sum_{a,x} X_{a|x} \left(\sum_{\vec{j}} \delta_{j_x,a} \mathcal{G}_{\vec{j}} - A_{a|x}^\eta \right). \end{aligned} \quad (1)$$

Notice that if we restrict our attention to dual variables that satisfy $Z_{\vec{j}} \geq 0$ and $\theta \geq 0$, then the last two terms of the first line of Eq. (1) are non-negative whenever the original (primal) variables satisfy the constraints of Eq. (4) of the main text. Similarly, the second line of Eq. (1) vanishes in this case, and we see that we have the inequality $\eta \leq \mathcal{L}$.

We can further restrict our attention to sets of dual variables which make the Lagrangian $\mathcal{L}$ independent of the primal variables $\{\mathcal{G}_{\vec{j}}\}_{\vec{j}}$ and η . To see this, we first factorize the Lagrangian,

$$\begin{aligned} \mathcal{L} = & \text{tr} \sum_{\vec{j}} \mathcal{G}_{\vec{j}} \left[Z_{\vec{j}} - \sum_{a,x} \delta_{j_x,a} X_{a|x} \right] \\ & + \eta \left[1 + \text{tr} \sum_{a,x} X_{a|x} \left(A_{a|x} - \text{tr} A_{a|x} \frac{\mathbb{1}}{d} \right) - \theta \right] \\ & + \theta + \frac{1}{d} \sum_{a,x} \text{tr} A_{a|x} \text{tr} X_{a|x}. \end{aligned} \quad (2)$$

Thus, if we consider only dual variables that satisfy the additional constraints

$$\theta = 1 + \text{tr} \sum_{a,x} X_{a|x} \left(A_{a|x} - \text{tr} A_{a|x} \frac{\mathbb{1}}{d} \right) \quad (3)$$

$$Z_{\vec{j}} = \sum_{a,x} \delta_{j_x,a} X_{a|x} \quad \forall \vec{j}, \quad (4)$$

then the two square brackets in Eq. (2) vanish, and we

are left with

$$\begin{aligned} \mathcal{L} = & \theta + \frac{1}{d} \sum_{a,x} \text{tr} A_{a|x} \text{tr} X_{a|x} \\ = & 1 + \text{tr} \sum_{a,x} X_{a|x} A_{a|x}, \end{aligned} \quad (5)$$

where the second line follows from the constraint (3). Thus, we arrive at an interesting situation, where for a particular set of dual variables, the Lagrangian $\mathcal{L}$, which by construction was bigger than η , is in fact independent of the primal variables. We can therefore obtain the tightest bound on η by minimizing the Lagrangian over this choice of dual variables, which is known as the dual problem, namely,

$$\min_{\{X_{a|x}\}_{a,x}, \{Z_{\vec{j}}\}_{\vec{j}}, \theta} 1 + \text{tr} \sum_{a,x} X_{a|x} A_{a|x} \quad (6)$$

$$\begin{aligned} \text{s.t.} \quad & \theta + \frac{1}{d} \sum_{a,x} \text{tr} A_{a|x} \text{tr} X_{a|x} = 1 + \text{tr} \sum_{a,x} X_{a|x} A_{a|x}, \\ & Z_{\vec{j}} = \sum_{a,x} \delta_{j_x,a} X_{a|x} \quad \forall \vec{j}, \\ & Z_{\vec{j}} \geq 0 \quad \forall \vec{j}, \quad \theta \geq 0. \end{aligned}$$

Additionally, we can use the first and second constraints to solve for $Z_{\vec{j}}$ and θ (formally, they are referred to as “slack variables”), which allows us to arrive at the final, simplified version of the dual problem which is given in Eq. (5) of the main text

$$\min_{\{X_{a|x}\}_{a,x}} 1 + \text{tr} \sum_{a,x} X_{a|x} A_{a|x} \quad (7)$$

$$\begin{aligned} \text{s.t.} \quad & 1 + \text{tr} \sum_{a,x} X_{a|x} A_{a|x} \geq \frac{1}{d} \sum_{a,x} \text{tr} A_{a|x} \text{tr} X_{a|x}, \\ & \sum_{a,x} \delta_{j_x,a} X_{a|x} \geq 0 \quad \forall \vec{j}. \end{aligned}$$

Finally, there is a theorem, known as the *strong duality theorem*, which is very powerful, and warrants the name “dual problem”: it states that if one can find a solution to either the primal problem or the dual problem that is strictly feasible (i.e., one can find positive-definite operators, rather than just positive-semidefinite operators that satisfy all the constraints), then the value of the dual problem is *equal* to the value of the primal problem. In the present case, taking $X_{a|x} = \mu \mathbb{1}$, for $\mu > 0$, gives a strictly feasible solution to the dual, and hence strong duality holds.

B. Ansatz solution

One of the key uses of the dual is that any feasible solution to the dual provides an upper bound on the primal problem. Let us make the following ansatz for the operators $X_{a|x}$, namely,

$$X_{a|x} = \alpha \mathbb{1} - \beta A_{a|x} \quad (8)$$

for some α and β that need to be determined. For this ansatz to satisfy the constraints of the dual (7), we must have

$$\begin{aligned} 1 + \text{tr} \sum_{a,x} X_{a|x} A_{a|x} - \frac{1}{d} \sum_{a,x} \text{tr} A_{a|x} \text{tr} X_{a|x} \\ = 1 - \beta \sum_{a,x} \left[\text{tr} A_{a|x}^2 - \frac{1}{d} (\text{tr} A_{a|x})^2 \right] \geq 0, \end{aligned} \quad (9)$$

and

$$\begin{aligned} \sum_{a,x} \delta_{jx,a} X_{a|x} &= \sum_{a,x} \delta_{jx,a} (\alpha \mathbb{1} - \beta A_{a|x}) \\ &= k\alpha \mathbb{1} - \beta \sum_x A_{jx|x} \geq 0. \end{aligned} \quad (10)$$

The first condition (9) is saturated if we pick

$$\beta = \frac{1}{\sum_{a,x} \left[\text{tr} A_{a|x}^2 - \frac{1}{d} (\text{tr} A_{a|x})^2 \right]}, \quad (11)$$

while, if we define

$$\lambda = \max_{\vec{j}} \left\| \sum_x A_{jx|x} \right\|_{\infty}, \quad (12)$$

which is the largest spectral radius of all the operators $\sum_x A_{jx|x}$, then

$$\alpha = \frac{\beta \lambda}{k} = \frac{\frac{\lambda}{k}}{\sum_{a,x} \left[\text{tr} A_{a|x}^2 - \frac{1}{d} (\text{tr} A_{a|x})^2 \right]} \quad (13)$$

satisfies the second condition (10), and cannot be improved. Substituting these values for α and β into the definition of $X_{a|x}$, we get that

$$X_{a|x} = \frac{\frac{\lambda}{k} \mathbb{1} - A_{a|x}}{\sum_{a',x'} \left[\text{tr} A_{a'|x'}^2 - \frac{1}{d} (\text{tr} A_{a'|x'})^2 \right]} \quad (14)$$

is a feasible point for the problem (7). Thus we finally arrive at the bound on the robustness

$$\eta \leq \frac{\lambda - \sum_{a,x} \left(\frac{\text{tr} A_{a|x}}{d} \right)^2}{\sum_{a,x} \left[\frac{\text{tr} A_{a|x}^2}{d} - \left(\frac{\text{tr} A_{a|x}}{d} \right)^2 \right]}. \quad (15)$$

C. Educated guess for a parent POVM

Here we show that if a parent POVM for the noisy POVMs is given by the operators $\mathcal{G}_{\vec{j}}$ defined by

$$\mathcal{G}_{\vec{j}} = \begin{cases} \Pi_{\vec{j}} & \text{if } \left\| \sum_x A_{jx|x} \right\|_{\infty} = \lambda \\ 0 & \text{otherwise,} \end{cases} \quad (16)$$

where $\Pi_{\vec{j}}$ is the projector on the eigenspace of $\sum_x A_{jx|x}$ associated with the eigenvalue λ , then the bound (15) is tight. In that sense, these operators are quite natural to try when the bound (15) is reached. This follows from Karush-Kuhn-Tucker conditions [1] but we derive it explicitly for pedagogical reasons.

Looking at Eq. (1), it is clear that *sufficient* conditions for the equality $\eta = \mathcal{L}$ to hold are

$$\theta = 0 \quad (17)$$

$$\text{tr} Z_{\vec{j}} \mathcal{G}_{\vec{j}} = 0 \quad \forall \vec{j} \quad (18)$$

$$\sum_{\vec{j}} \delta_{jx,a} \mathcal{G}_{\vec{j}} = A_{a|x}^{\eta} \quad \forall a, x. \quad (19)$$

It is easy to see that choosing the ansatz (14) together with the operators $\mathcal{G}_{\vec{j}}$ defined in Eq. (16) fulfills (17) and (18). Importantly, the definition (16) is designed to make (18) true. Therefore, if the operators defined in Eq. (16) satisfy the constraint (19), that is, form a parent POVM for $\{A_{a|1}^{\eta}\}_a, \dots, \{A_{a|k}^{\eta}\}_a$, then all three conditions are satisfied so that $\eta = \mathcal{L}$. Since equality between primal and dual objective functions can only be true at their optimum, we know that the parent POVM $\mathcal{G}_{\vec{j}}$ is, in that case, associated to the noise robustness η^* .

It is remarkable that this educated guess of $\mathcal{G}_{\vec{j}}$ works for some sets of MUB, as stated in the main text and proven in Sec. IV. We did not find any other set of measurement with this property. Moreover, we found a set of MUB for which the bound (15) is tight without the operators (16) being a parent POVM for the noisy MUB (see Sec. III for details).

II. QUADRUPLTS OF MUB IN DIMENSION SIX

In dimension six, the number of MUB one can construct is unknown [2], even though there are now a strong belief that no more than three can exist. In this section, we nonetheless assume that four MUB do exist in dimension six. We show that we can still derive a bound on λ defined in Eq. (7) of the main text. The idea is to use the mutual unbiasedness (Eq. (1) of the main text) to get some information on the characteristic polynomial of $S_{\vec{j}}$.

Given k MUB $\{|\varphi_j^1\rangle\}_j, \dots, \{|\varphi_j^k\rangle\}_j$ and the corresponding $S_{\vec{j}} = \sum_x |\varphi_{jx}^x\rangle \langle \varphi_{jx}^x|$, it is straightforward to check the

following equalities

$$\text{tr } S_j = k \quad (20)$$

$$\text{tr } S_j^2 = k \left(\frac{k-1}{d} + 1 \right) \quad (21)$$

$$\text{tr } S_j^3 = k \left(3 \frac{k-1}{d} + 1 \right) + \sigma_j^{(3)} \quad (22)$$

$$\text{tr } S_j^4 = k \left[6 \frac{k-1}{d} + 1 + \left(\frac{k-1}{d} \right)^2 \right] + 4\sigma_j^{(3)} + \sigma_j^{(4)}, \quad (23)$$

where we have defined

$$\sigma_j^{(3)} = \sum_{\substack{a,b,c=1 \\ \neq}}^k \langle \varphi_{ja}^a | \varphi_{jb}^b \rangle \langle \varphi_{jb}^b | \varphi_{jc}^c \rangle \langle \varphi_{jc}^c | \varphi_{ja}^a \rangle \quad (24)$$

and

$$\sigma_j^{(4)} = \sum_{\substack{a,b,c,d=1 \\ \neq}}^k \langle \varphi_{ja}^a | \varphi_{jb}^b \rangle \langle \varphi_{jb}^b | \varphi_{jc}^c \rangle \langle \varphi_{jc}^c | \varphi_{jd}^d \rangle \langle \varphi_{jd}^d | \varphi_{ja}^a \rangle \quad (25)$$

with the sums running over indices that take different values (parwise). Thanks to Newton's identities, we can express the characteristic polynomial of S_j using the traces of successive powers of S_j . When $\text{rank}(S_j) = 4$, it reads

$$\begin{aligned} X^{d-4} & \left\{ X^4 - kX^3 + \frac{k(k-1)}{2} \left(1 - \frac{1}{d} \right) X^2 \right. \\ & - \left[\frac{k(k-1)(k-2)}{6} \left(1 - \frac{3}{d} \right) + \frac{\sigma_j^{(3)}}{3} \right] X \\ & + \frac{k(k-1)(k-2)(k-3)}{24} \left(1 - \frac{6}{d} \right) \\ & \left. + \frac{k(k-1)^2(k-2)}{8d^2} + (k-3) \frac{\sigma_j^{(3)}}{3} + \frac{\sigma_j^{(4)}}{4} \right\}. \end{aligned} \quad (26)$$

In the specific case of $k = 4$ and $d = 6$, this gives a simple expression for the characteristic polynomial of S_j , namely,

$$\frac{1}{4} \left[(2X^2 - 4X + 1)^2 - \sigma_4 \right] - \frac{1}{3} \sigma_3 (X - 1). \quad (27)$$

Then it can be seen that its maximum root is an increasing function of both $\sigma_j^{(3)}$ and $\sigma_j^{(4)}$ so that we get an upper bound on λ by taking their trivial upper bounds $|\sigma_j^{(3)}| \leq 24/d\sqrt{d}$ and $|\sigma_j^{(4)}| \leq 24/d^2$ achieved by the triangle inequality. This bound is approximately $x \leq 2.183$ so that we have $\eta_{\text{up}} \leq 0.4550$.

Then we can also compute the value of η_{low} since it does not require the explicit form of the MUB. This leads to the following bounds for the value of η^* in the case of hypothetical quadruplets of MUB in dimension six

$$0.4175 \leq \eta^* \leq 0.4550. \quad (28)$$

TABLE I. Analytical forms of the noise robustness η^* of k MUB. The cases $k = 2$ or $d = 2$ are not given since they were already known (see, e.g., [3]). Multiple values in one cell are due to the existence of inequivalent sets. In dimension nine, for $k = 4$ and $k = 6$, the value concerns only one of the three inequivalent sets (see Table II of the main text). Importantly, as explained in the main text, in some cases the equality $\eta^* = \eta_{\text{up}}$ is only valid up to numerical precision.

$k \setminus d$	3	4	5	8	9
3	$\frac{\cos(\pi/18)}{\sqrt{3}}$		$\frac{1+\sqrt{5}}{6}$ $\frac{13-\sqrt{5}+\sqrt{30(5+\sqrt{5})}}{48}$		$\frac{1}{2}$ $\frac{1+\cos(\pi/9)}{4}$
4	$\frac{1+3\sqrt{5}}{16}$	$\frac{1}{2}$			$\frac{8+3\sqrt{3}}{32}$
5		$\frac{3+2\sqrt{3}}{15}$			
6					$\frac{3+\sqrt{7}}{16}$
9				$\frac{3+2\sqrt{3}}{21}$	

All this procedure also works for quadruplets of MUB in other dimensions in which only three MUB are known to exist. For example, in dimension ten, it gives $0.3864 \leq \eta^* \leq 0.4213$.

III. ANALYTICAL VALUES

In Table I of the main text, we give most of the analytical values we found for η^* . Here we add further ones in Table I together with explanations on their origins.

When the operators $\mathcal{G}_j$ defined in Eq. (11) of the main text form a valid parent POVM for noisy versions of $\{A_{a|1}\}_a, \dots, \{A_{a|k}\}_a$, we know from Sec. I that the bound η_{up} is tight. This provides us with a simple sufficient criterion to get analytical values for η^* . For $k = d$ and $k = d + 1$, as claimed in the main text and proven in Sec. IV, the validity of this condition holds analytically. Moreover, for one of the triplets in dimension five, namely, the one giving rise to $\eta_{\text{up}} = (1 + \sqrt{5})/6$, we could check it explicitly. However, in the other cases, it could only be checked numerically thus the equality $\eta^* = \eta_{\text{up}}$ is only valid up to numerical precision.

Similarly, we could also compare the numerical output η^* of the SDP (5) with the value of the upper bound η_{up} . Other instances of the equality $\eta^* = \eta_{\text{up}}$ were found by this method, up to numerical precision. For example, for triplets in dimension nine, the POVM (11) is not a parent POVM though the upper bound (9) is reached, up to numerical precision.

In Table I, we do not give any value, e.g., for $k = 5$ and $d = 5$ though we claim in the main text that our method gives an analytical result in that case. This is because in that case, η^* is a root of a polynomial which we cannot solve explicitly.

Eventually, there are cases in which an analytic form is known but not reproduced here because it is too heavy. For example, for $k = 7$ and $d = 7$, $\eta^* \approx 0.3685$ is the

largest root of $56X^3 - 28X^2 + 1$, which can be solved explicitly.

IV. TIGHTNESS OF THE UPPER BOUND FOR SPECIFIC SETS OF MUB

A. A sequence converging to $\mathcal{G}_{\vec{j}}$

The expression of $\mathcal{G}_{\vec{j}}$, though naturally emerging from the procedure of Sec. I, is very difficult to manipulate since the expression of the eigenvectors of $S_{\vec{j}}$ is unknown in general. Here we express $\mathcal{G}_{\vec{j}}$ as the limit of the sequence

$$\mathcal{G}_{\vec{j}}^{(n)} = \left(\frac{S_{\vec{j}}}{\lambda} \right)^n, \quad (29)$$

where we recall from the definition (7) of the main text that $\lambda = \max_{\vec{j}} \|S_{\vec{j}}\|_{\infty}$. This can easily be seen by writing $S_{\vec{j}}/\lambda$ in a diagonal form.

The point of expressing $\mathcal{G}_{\vec{j}}$ like this is that the elements $\mathcal{G}_{\vec{j}}^{(n)}$ of the sequence are simpler to handle. Thus we will prove our results on each $\mathcal{G}_{\vec{j}}^{(n)}$ so that it will also hold for the limit by continuity.

Interestingly, for $n \leq 4$, without any other assumption than the unbiasedness of the bases, $\mathcal{G}_{\vec{j}}^{(n)}$ can be tediously proven to form, up to normalization, a parent POVM for $\{A_{a|1}^{\eta^{(n)}}\}_a, \dots, \{A_{a|k}^{\eta^{(n)}}\}_a$ with $\eta^{(1)} = 1/k$, $\eta^{(2)} = \frac{d+2(k-1)}{k(d+k-1)}$, $\eta^{(3)} = \frac{d^2+5(k-1)d+3(k-1)(k-2)}{k[d^2+3(k-1)d+(k-1)(k-2)]}$, and $\eta^{(4)} = \frac{d^3+9(k-1)d^2+2(k-1)(7k-13)d+4(k-1)(k-2)(k-3)}{k[d^3+6(k-1)d^2+(k-1)(6k-11)d+(k-1)(k-2)(k-3)]}$.

With some more effort, it is also possible to extend this to any n using a specific form for the MUB and the assumption $k = d$ or $k = d + 1$. This is the goal of the next subsections.

B. Odd prime power

In Ref. [4] the following bases together with the computational one are proven to form a complete set of MUB in dimension $d = p^r$ where p is an odd prime number

$$|\varphi_a^x\rangle = \frac{1}{\sqrt{d}} \sum_{l \in \mathbb{F}_d} \omega_p^{\text{Tr}(xl^2 + al)} |l\rangle, \quad (30)$$

where the basis label x and the vector label a are in the Galois field $\mathbb{F}_d$ with d elements, $\{|l\rangle\}_{l \in \mathbb{F}_d}$ is the computational basis, ω_p is a p th root of the unity $\exp(2i\pi/p)$, and Tr is the trace on $\mathbb{F}_d$, defined by $\text{Tr } a = a + a^p + \dots + a^{p^{r-1}}$ so that it belongs to $\mathbb{F}_p = \mathbb{Z}_p$.

We recall some basic notions about Galois fields. For further information, we refer the interested reader to Refs [5, 6]. A Galois field is a finite set with two internal operations that have basically the same properties as the

usual addition and multiplication: associativity, commutativity, existence of a unit element and of an inverse for all elements, and distributivity. Having all these properties is very restrictive and only finite sets with a number of element $d = p^r$ with p prime and r positive integer are able to satisfy them. This is why the construction only works in prime power dimensions. The trace on Galois fields is simply a map from the abstract field to the set $0, \dots, p-1$ such that $\text{Tr}(1) = 1$, $\text{Tr}(a+b) = \text{Tr } a + \text{Tr } b$, and $\text{Tr}(ab) = (\text{Tr } a)(\text{Tr } b)$.

To keep notations simple, we will only consider in the following the case $k = d$ and all these MUB except the computational basis. By treating the computational basis separately, the proof can be straightforwardly adapted to the other cases, namely, the other subsets of d MUB and the complete set of $d+1$ MUB.

1. Normalization

Here we prove that for all n , $\mathcal{G}_{\vec{j}}^{(n)}$ is a POVM, up to normalization. Since the positivity is immediate from the definition (29), we are left to show that these operators sum up to 1, up to normalization. Interestingly the proof will also turn out to be valid for any k and any subset of MUB. By definition (7) of the main text we have

$$\sum_{\vec{j}[k] \in \mathbb{F}_d} S_{\vec{j}}^n = \sum_{\vec{j}} \left(\sum_{x \in \mathbb{F}_d} |\varphi_{j_x}^x\rangle \langle \varphi_{j_x}^x| \right)^n \quad (31)$$

$$= \sum_{\vec{j}} \sum_{\vec{x}[n]} |\varphi_{j_{x_1}}^{x_1}\rangle \langle \varphi_{j_{x_1}}^{x_1}| \dots |\varphi_{j_{x_n}}^{x_n}\rangle \langle \varphi_{j_{x_n}}^{x_n}| \quad (32)$$

Then we choose a basis $\alpha \in \mathbb{F}_d$ and we introduce a closure relation $\sum_{l \in \mathbb{F}_d} |\varphi_l^\alpha\rangle \langle \varphi_l^\alpha|$ between all factors. This trick is equivalent to projecting all vectors is the basis $\{|\varphi_l^\alpha\rangle\}_l$. Each term of (32) then becomes

$$\sum_{\vec{l}[n+1] \in \mathbb{F}_d} |\varphi_{l_1}^\alpha\rangle \langle \varphi_{l_1}^\alpha| \varphi_{j_{x_1}}^{x_1}\rangle \langle \varphi_{j_{x_1}}^{x_1}| \dots |\varphi_{j_{x_n}}^{x_n}\rangle \langle \varphi_{j_{x_n}}^{x_n}| \varphi_{l_{n+1}}^\alpha\rangle \langle \varphi_{l_{n+1}}^\alpha|, \quad (33)$$

so that we can regroup the scalar products $\langle \varphi_{l_i}^\alpha | \varphi_{j_{x_i}}^{x_i} \rangle \langle \varphi_{j_{x_i}}^{x_i} | \varphi_{l_{i+1}}^\alpha \rangle$ depending on the value of x_i to perform the sum over $\vec{j}$, namely, the right hand side of (32) is

$$\sum_{\vec{x}, \vec{l}} \left(\prod_{m \in \mathbb{F}_d} \sum_{j_m} \prod_{\substack{i \\ x_i = m}} \langle \varphi_{l_i}^\alpha | \varphi_{j_{x_i}}^{x_i} \rangle \langle \varphi_{j_{x_i}}^{x_i} | \varphi_{l_{i+1}}^\alpha \rangle \right) |\varphi_{l_1}^\alpha\rangle \langle \varphi_{l_{n+1}}^\alpha|. \quad (34)$$

Using the definition (30) and quadratic Gauss sums on Galois fields [6], when $x_i \neq \alpha$, we have

$$\langle \varphi_{l_i}^\alpha | \varphi_{j_{x_i}}^{x_i} \rangle \langle \varphi_{j_{x_i}}^{x_i} | \varphi_{l_{i+1}}^\alpha \rangle = \frac{1}{d} \omega_p^{\text{Tr}(\mu[-(l_i^2 - l_{i+1}^2) + 2j_x(l_i - l_{i+1})])}, \quad (35)$$

where $\mu = 2^{-2}(x_i - \alpha)^{-1}$. Hence, thanks to the geometric sum $\sum_{x \in \mathbb{F}_d} \omega_p^{\text{Tr}(ax)} = d\delta_a^0$, this becomes

$$\sum_{j_m} \prod_{x_i=m} \langle \varphi_{l_i}^\alpha | \varphi_{j_{x_i}}^{x_i} \rangle \langle \varphi_{j_{x_i}}^{x_i} | \varphi_{l_{i+1}}^\alpha \rangle \propto \delta_{\sum_{x_i=m} (l_i - l_{i+1})}^0 \quad (36)$$

where δ_x^y is the Kronecker delta in $\mathbb{F}_d$ which is 1 if $x = y$ and 0 otherwise. Combining all this, we notice that

$$l_1 - l_{n+1} = \sum_{m \in \mathbb{F}_d} \sum_{x_i=m} (l_i - l_{i+1}) = 0 \quad (37)$$

so that $\sum_{\vec{j}} \mathcal{G}_{\vec{j}}^{(n)}$ is diagonal in the basis $\{|\varphi_l^\alpha\rangle\}_l$. Since this is true for all α , it seems reasonable that $\sum_{\vec{j}} \mathcal{G}_{\vec{j}}^{(n)}$ is a multiple of $\mathbb{1}$. The following lemma formalizes this idea.

Let M be an operator which is diagonal in two MUB denoted by $\{|\varphi_i^1\rangle\}_i$ and $\{|\varphi_j^2\rangle\}_j$, i.e., $M = \sum_i m_i^{(1)} |\varphi_i^1\rangle \langle \varphi_i^1| = \sum_j m_j^{(2)} |\varphi_j^2\rangle \langle \varphi_j^2|$. Then by decomposing $|\varphi_j^2\rangle$ in $|\varphi_i^1\rangle$ we get $m_i^{(1)} = \sum_l \langle \varphi_i^1 | \varphi_l^2 \rangle \langle \varphi_l^2 | \varphi_i^1 \rangle m_l^{(2)}$ which is constant since the bases are mutually unbiased.

Hence the normalization of the previously introduced $\mathcal{G}_{\vec{j}}^{(n)}$ is achievable. Thus, by going to the limit, $\mathcal{G}_{\vec{j}}$ can also be normalized. Note that the proof provided here is valid for all k , which means that all the operators defined in (11) of the main text are always POVMs, up to normalization.

2. Marginals

Here we compute the marginals of $\mathcal{G}_{\vec{j}}^{(n)}$, i.e., its sum over $\vec{j}[k]$ with j_α fixed to γ , where we set α and γ are in $\mathbb{F}_d$. The goal is to show that these marginals are of the form $\eta |\varphi_\gamma^\alpha\rangle \langle \varphi_\gamma^\alpha| + (1 - \eta) \mathbb{1}/d$, up to normalization. The first steps are essentially the same as in the previous section: brute-force development of the power into many sums, injection of many closure relations to write the marginal in the basis $\{|\varphi_x^\alpha\rangle\}_x$, explicit evaluation of all the scalar products involved to get rid of the sum over $\vec{j}[k]$, and combination of the resulting Kronecker deltas to get the diagonality of the marginal. Then, by relabeling the indices $\vec{x}$ so that cumbersome coefficients vanish, this becomes

$$\sum_{\vec{j}[k]} \delta_{j_\alpha, \gamma} S_{\vec{j}}^n = \sum_{l_1 \in \mathbb{F}_d} \left(\sum_{\vec{x}[n] \in \mathbb{F}_d} \sum_{l_2 \dots l_n} c_{\vec{x}, \vec{l}} \right) |\varphi_{l_1}^\alpha\rangle \langle \varphi_{l_1}^\alpha| \quad (38)$$

where the coefficients $c_{\vec{x}, \vec{l}}$ are defined to be

$$\prod_{m \in \mathbb{F}_d} \delta_{\sum_{x_i=m} (l_i - l_{i+1})}^0 \prod_i \delta_{l_i}^\gamma \delta_{l_{i+1}}^\gamma \prod_{i=1}^n \omega_p^{\text{Tr}(x_i(l_i^2 - l_{i+1}^2))}. \quad (39)$$

At this stage, we introduce, for a given $\vec{x}$, the partition $X_1, \dots, X_\tau$ of $[1, n]$ that naturally emerges when grouping the x_i by their values. With this, the sum over $\vec{x}$ can be decomposed into a sum over the partitions of $[1, n]$ and the sums over the different values taken by the x_i in the different sets of each partition. Let $X_1, \dots, X_\tau$ be a fixed partition of $[1, n]$. We denote by α_ρ the common value of x_i for $i \in X_\rho$. Up to some manipulation, the computation of the coefficients of the diagonal expansion (38) leads to the evaluation of the following sum

$$\sigma(\vec{\beta}[\epsilon]) = \sum_{\alpha_1 \in \mathbb{F}_d^*} \omega_p^{\text{Tr}(\alpha_1 \beta_1)} \sum_{\substack{\alpha_2 \in \mathbb{F}_d^* \\ \alpha_2 \neq \alpha_1}} \omega_p^{\text{Tr}(\alpha_2 \beta_2)} \dots \sum_{\substack{\alpha_\epsilon \in \mathbb{F}_d^* \\ \forall \rho < \epsilon, \alpha_\epsilon \neq \alpha_\rho}} \omega_p^{\text{Tr}(\alpha_\epsilon \beta_\epsilon)} \quad (40)$$

where $\beta_\rho = \sum_{i \in X_\rho} (l_i^2 - l_{i+1}^2)$. This is where the assumption on k becomes a *sufficient* condition to continue: if $k = d$ or $k = d + 1$, the geometric sum $\sum_{\alpha_\epsilon \in \mathbb{F}_d^*} \omega_p^{\text{Tr}(\alpha_\epsilon \beta_\epsilon)} = d\delta_{\beta_\epsilon}^0 - 1$ can be used to get a recursive expression of $\sigma(\vec{\beta}[\epsilon])$. For example, when $k = d$, it reads

$$\sigma(\vec{\beta}[\epsilon]) = (d\delta_{\beta_\epsilon}^0 - 1) \sigma(\vec{\beta}[\epsilon - 1]) - \sum_{\rho=1}^{\epsilon-1} \sigma(\beta_1, \dots, \beta_\rho + \beta_\epsilon, \dots, \beta_{\epsilon-1}), \quad (41)$$

together with the initialization $\sigma(\beta_1) = d\delta_{\beta_1}^0 - 1$. Using this, the coefficients of the diagonal expansion (38) become a complicated sum of cardinals of sets. Fortunately, these cardinals are invariant when $l_1 \neq \gamma$ varies so that it can be eventually seen that

$$\begin{aligned} \sum_{\vec{j}[k]} \delta_{j_\alpha, \gamma} S_{\vec{j}}^n &= r |\varphi_\gamma^\alpha\rangle \langle \varphi_\gamma^\alpha| + s \sum_{l_1 \neq \gamma} |\varphi_{l_1}^\alpha\rangle \langle \varphi_{l_1}^\alpha| \\ &= (r - s) |\varphi_\gamma^\alpha\rangle \langle \varphi_\gamma^\alpha| + s \mathbb{1}, \end{aligned} \quad (42)$$

with $r > s$. The desired behavior, namely, that $\mathcal{G}_{\vec{j}}$ forms a parent POVM for noisy versions of our MUB measurements, follows.

C. Even prime power

In Ref. [4] the following bases together with the computational one are proven to form a complete set of MUB in dimension $d = 2^r$

$$|\varphi_a^x\rangle = \frac{1}{\sqrt{d}} \sum_{l \in \mathcal{T}_r} i^{\text{Tr}[(x+2a)l]} |l\rangle, \quad (43)$$

where the basis label x and the vector label a are in the Teichmüller set $\mathcal{T}_r$ of the Galois ring $\text{GR}(4, r)$, $\{|l\rangle\}_{l \in \mathcal{T}_r}$ is the computational basis, and Tr is the trace on $\text{GR}(4, r)$. We refer to Ref. [7] for an introduction to Galois rings.

Then, by using Lemma 3 of Ref. [8] instead of quadratic Gauss sums, we can derive the equivalent of Eq. (35)

when $x_i \neq \alpha$, namely,

$$\langle \varphi_{l_i}^\alpha | \varphi_{j_{x_i}}^{x_i} \rangle \langle \varphi_{j_{x_i}}^{x_i} | \varphi_{l_{i+1}}^\alpha \rangle = \frac{1}{d} \text{Tr}[t_i^{-2}(x_i - \alpha + 2j_{x_i})(l_{i+1} - l_i)], \quad (44)$$

where $x_i - \alpha = t_i + 2u_i$ with $t_i \in \mathcal{T}_r^*$ and $u_i \in \mathcal{T}_r$. Then the proof is the same as in the odd case. It is even simpler since there is no quadratic term in l_i in the exponent of Eq. (44) so that the third product in Eq. (39) disappears.

Thus the same result follows, namely, $\mathcal{G}_{\vec{j}}$ is a parent POVM for noisy versions of our MUB measurements.

V. LOWER BOUND FOR MUB

Here we give more details about the construction of the lower bound proposed in the main text, together with numerical values for $d \leq 7$ in Table II.

We recall the procedure. Using only the property of the bases to be mutually unbiased (see Eq. (1) of the main text), we can prove the operators $\mathcal{G}_{\vec{j}}$ defined in Eq. (11) of the main text to form a parent POVM for noisy versions of our MUB measurements. The corresponding noise parameter η_k given in Eq. (14) of the main text has some freedom intentionally introduced via the parameters α_i , $2 \leq i \leq k$. As the construction provides, by Eq. (4) of the main text, a lower bound to η^* , we aim at maximizing the value of the function $\eta_k(\alpha_2, \dots, \alpha_k)$.

The iterative definition of $\eta_k(\eta_{k-1}, \alpha_k)$ (Eq. (14) of the main text) is increasing with η_{k-1} since we choose positive α_k . Thus the optimization of the different parameters can clearly be made successively and independently. We make it even more explicit in the following. First we optimize $\eta_2(\alpha_2)$ over the single parameter α_2 . By a simple derivative computation, we get the argument $\alpha_2^* = 1$ for which the maximum η_2^* of η_2 is reached. Then we optimize $\eta_3(\eta_2^*, \alpha_3)$ over the single parameter α_3 . This gives an argument

$$\alpha_3^* = \frac{\sqrt{5d + 12\sqrt{d} + 8} - \sqrt{d}}{2(\sqrt{d} + 2)} \quad (45)$$

giving rise to the maximum η_3^* which we plug into the next step, and so on.

VI. IMPLICATIONS FOR EINSTEIN-PODOLSKY-ROSEN STEERING

The results presented in the main text also have implications for EPR steering, due to the intimate relation that exists with joint measurability [9–11]. In an EPR steering scenario, two parties called Alice and Bob share a quantum state, onto which Alice performs measurements, in order to “steer” the state of Bob’s system. In particular, if the shared state is ρ^{AB} , and Alice performs POVMs from the family $\{\{A_{a|x}\}_a\}_x$, then the sub-normalized states prepared for Bob are given by

TABLE II. The lower bound η_{low} on the noise robustness η^* we get by deriving an explicit parent POVM for k MUB in dimension d using only the mutual unbiasedness of the bases.

$k \setminus d$	2	3	4	5	6	7
2	0.7071	0.6830	0.6667	0.6545	0.6449	0.6371
3	0.5774	0.5468	0.5263	0.5113	0.4996	0.4902
4		0.4672	0.4455	0.4297	0.4175	0.4076
5			0.3918	0.3758	0.3636	0.3537
6				0.3371	0.3250	0.3153
7					0.2958	0.2863
8						0.2634

$\sigma_{a|x} = \text{tr}_A[(A_{a|x} \otimes \mathbb{1})\rho^{AB}]$, which occur with probability $p(a|x) = \text{tr} \sigma_{a|x}$. The collection of sub-normalized states $\{\{\sigma_{a|x}\}_a\}_x$ is known as an assemblage, and contains all of the observable data in an EPR steering scenario. Steering is said to be demonstrated if this assemblage cannot be reproduced by a local-hidden-state (LHS) model, which takes the form of a collection of sub-normalized states $\{\sigma_{\vec{j}}\}_{\vec{j}}$ such that with probability $\text{tr} \sigma_{\vec{j}}$, the state $\sigma_{\vec{j}}$ is sent to Bob and $\vec{j}$ is sent to Alice, who will announce $a = j_x$ as the outcome for the measurement labelled x . The LHS model will reproduce the assemblage $\{\{\sigma_{a|x}\}_a\}_x$ if $\sum_{\vec{j}} \delta_{j_x, a} \sigma_{\vec{j}} = \sigma_{a|x}$.

In Refs [9, 11] it was shown that joint measurability and EPR steering are equivalent in the following sense: any set of non-jointly-measurable measurements $\{\{A_{a|x}\}_a\}_x$, when used by Alice on any pure full-Schmidt-rank entangled state, $|\psi\rangle = \sum_i \sqrt{\lambda_i} |i\rangle |i\rangle$, where $\lambda_i > 0$ for all i , produces an assemblage $\{\{\sigma_{a|x}\}_a\}_x$ that demonstrates EPR steering. That is, for pure full-Schmidt-rank states, it is necessary and sufficient to use non-jointly-measurable measurements to demonstrate EPR steering.

For any quantum state $|\psi\rangle$, a noisy version is given by $\rho_\psi^w = w|\psi\rangle\langle\psi| + (1-w)\mathbb{1}/d \otimes \text{tr}_A |\psi\rangle\langle\psi|$. When $|\psi\rangle = |\Phi^+\rangle = \sum_i |i\rangle |i\rangle / \sqrt{d}$ is the maximally entangled state, then $\rho_{\Phi^+}^w$ is the isotropic state. A basic question about EPR steering is to determine the critical noise w^* below which a state ρ^w becomes unsteerable (has an LHS model for all measurements) and above which it demonstrates steering.

The present analysis allows us to obtain the upper bound $w^* \leq \eta^*$. In particular, for $\eta > \eta^*$ then projective MUB measurements are not jointly measurable, hence when Alice uses them on any full-Schmidt-rank state $|\psi\rangle$ the resulting assemblage demonstrates steering. However, the identity

$$\text{tr}_A[(A_{a|x}^\eta \otimes \mathbb{1})|\psi\rangle\langle\psi|] = \text{tr}_A[(A_{a|x} \otimes \mathbb{1})\rho_\psi^\eta] \quad (46)$$

shows that the exact same assemblage would arise from noiseless measurements on the noisy state ρ_ψ^η , which therefore also demonstrates steering. Thus the values

given in Table I of the main text are bounds on the robustness of pure entangled states above which EPR steering can be demonstrated by using MUB measurements.

VII. OPTIMALITY OF MUB' NOISE ROBUSTNESS IN THE QUBIT CASE

In this section we show that, among pairs (resp., triplets) of two-outcome unbiased qubit POVMs, the projective POVMs of two (resp., three) MUB reaches the minimal noise parameter $\eta^* = 1/\sqrt{2}$ (resp., $\eta^* = 1/\sqrt{3}$). For pairs of POVM, this is actually already known [12], but the proof is given to make the triplet-case clearer. A two-outcome unbiased qubit POVM A_i is given by

$$A_{\mu_i|i} = \frac{\mathbb{1} + \mu_i \vec{a}_i \cdot \vec{\sigma}}{2}, \quad (47)$$

where μ_i can take the value $+1$ or -1 . It is said to be *unbiased* because $\text{tr } A_{+1|i} = \text{tr } A_{-1|i} = 1/2$.

For pairs of POVMs, we introduce the following operators

$$G_{\mu_1\mu_2} = \frac{(1 + \mu_1\mu_2 Z)\mathbb{1} + \eta(\mu_1\vec{a}_1 + \mu_2\vec{a}_2) \cdot \vec{\sigma}}{4}, \quad (48)$$

where $Z = 1 - \eta\|\vec{a}_1 - \vec{a}_2\|$. They obviously form a parent POVM of A_1^η and A_2^η as soon as they are all positive, which is the case whenever

$$\eta \leq \eta_2^* = \frac{2}{\|\vec{a}_1 + \vec{a}_2\| + \|\vec{a}_1 - \vec{a}_2\|}. \quad (49)$$

This is a well-known condition which is also sufficient [3]. Through the Cauchy-Schwarz inequality, we get $\eta_2^* \geq 1/\sqrt{2}$ with equality if and only if the two POVMs are projective MUB measurements, i.e., $\|\vec{a}_i\| = 1$ and $\vec{a}_1 \cdot \vec{a}_2 = 0$.

For triplets of POVMs, taking inspiration from Refs [13, 15], we introduce

$$G_{\mu_1\mu_2\mu_3} = \frac{1}{8} \left[(1 + \mu_2\mu_3 Z_1 + \mu_3\mu_1 Z_2 + \mu_1\mu_2 Z_3)\mathbb{1} + \eta(\mu_1\vec{a}_1 + \mu_2\vec{a}_2 + \mu_3\vec{a}_3) \cdot \vec{\sigma} \right], \quad (50)$$

where

$$Z_1 = 1 - \eta \frac{\|\vec{a}_2 - \vec{a}_1 - \vec{a}_3\| + \|\vec{a}_3 - \vec{a}_1 - \vec{a}_2\|}{2} \quad (51)$$

$$Z_2 = 1 - \eta \frac{\|\vec{a}_1 - \vec{a}_2 - \vec{a}_3\| + \|\vec{a}_3 - \vec{a}_1 - \vec{a}_2\|}{2} \quad (52)$$

$$Z_3 = 1 - \eta \frac{\|\vec{a}_1 - \vec{a}_2 - \vec{a}_3\| + \|\vec{a}_2 - \vec{a}_1 - \vec{a}_3\|}{2}. \quad (53)$$

They obviously form a parent POVM of A_1^η , A_2^η and A_3^η as soon as they are all positive, which is the case whenever

$$\eta \leq \eta_3^* = \frac{4}{\Sigma}, \quad (54)$$

where

$$\Sigma = \|\vec{a}_1 + \vec{a}_2 + \vec{a}_3\| + \|\vec{a}_1 - \vec{a}_2 - \vec{a}_3\| + \|\vec{a}_2 - \vec{a}_1 - \vec{a}_3\| + \|\vec{a}_3 - \vec{a}_1 - \vec{a}_2\|. \quad (55)$$

This condition is only necessary [15] but it will be enough for our needs. Indeed, through the Cauchy-Schwarz inequality, we get $\eta_3^* \geq 1/\sqrt{3}$ with equality if and only if the two POVMs are projective MUB measurements, i.e., $\|\vec{a}_i\| = 1$ and $\vec{a}_i \cdot \vec{a}_j = 0$ for $i \neq j$.

In the case of pairs of POVMs, the result can even be extended to any pair of two-outcome qubit POVM. In Ref. [12], the problem of joint measurement of any pair of two-outcome qubit POVM is indeed completely solved. More specifically, an inequality involving the POVM parameters is given which is necessary and sufficient for their joint measurability. For our needs, the sufficient inequality $F_x^2 + F_y^2 \geq 1$ mentioned in Ref. [12] is enough to ensure that the most robust to white noise pairs of two-outcome qubit POVM are projective measurements on two MUB.

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